

Datacenter load, demand response, and resource adequacy

PJM and ERCOT · ReEDS 2032 · PRAS

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Datacenter load, demand response, and resource adequacy — brief results

PJM and ERCOT, ReEDS model year 2032 · PRAS · 15 weather years (2007-2021) × 1000

Monte Carlo samples Input set: `neue1` (a more stressed capacity mix than the earlier runs — see note). Prepared 2026-07-23 · reproducible via `sbatch slurm/run_all.sh`, see [README.md](#)

Note on inputs. These results use the `neue1` PRAS systems (`{PJM,ERCOT}_PRAS_2032*_neue1*.pras`). They are far more capacity-constrained than the first-round systems: PJM's no-DR NEUE is ~330× higher than before and ERCOT — previously zero — now has the larger reliability problem of the two. An earlier version of this document described the first-round systems, in which ERCOT had no unserved energy and PJM's was a single-region artifact. That story does not carry over; this is a different, much tighter system.

What was done

ReEDS built a 2032 grid under the **default** demand forecast — every generator, battery and transmission line. That fleet was then **frozen**.

PRAS took the frozen fleet and simulated the year hour by hour, 8760 hours × 15 weather years, 1000 times over, randomly failing generators at their forced-outage rates. Each hour it asks whether generation plus imports could serve load; when they could not, it records the shortfall. **NEUE** (normalized expected unserved energy, in parts per million) is that shortfall as a share of total load.

That simulation was run twice — once with normal load, once with datacenter load added on top — **using the same power plants both times**. Then demand response was layered on: each

region received a DR device sized as a fraction of *its own added datacenter load*, swept from 5 % to 100 %, in two flavors:

- **shift** — load moved to a later hour and repaid within the payback window; available 24/7.
- **shed** — load dropped and never repaid; available only in a fixed daily window.

Because the buildout never changes, any difference in NEUE is attributable to the added load and to the DR — not to different generation. The question being answered is deliberately narrow: **can flexible datacenter load substitute for generation that was never built?**

92 PRAS runs in total: 68 PJM (shed + shift × {4,8,16} h × 11 fractions), 24 ERCOT (shift only × {4,8} h × 11 fractions).

Headline results

1. Both systems are heavily stressed; ERCOT is now the worse of the two

	base (no DC)	+ datacenter load, no DR	best DR case
PJM	1.05 ppm	59.8 ppm (1,003,700 MWh)	20.6 ppm — shed_16h @ 100 %
ERCOT	1.00 ppm	72.1 ppm (804,376 MWh)	56.2 ppm — shift_8h @ 100 %

Adding datacenter load raises NEUE **61× in PJM** and **79× in ERCOT**. These are large reliability shortfalls in absolute terms, not the marginal signals of the earlier runs. Both systems are genuinely short of capacity once the datacenter load arrives.

2. EUE lands where the datacenter load lands — the opposite of the earlier runs

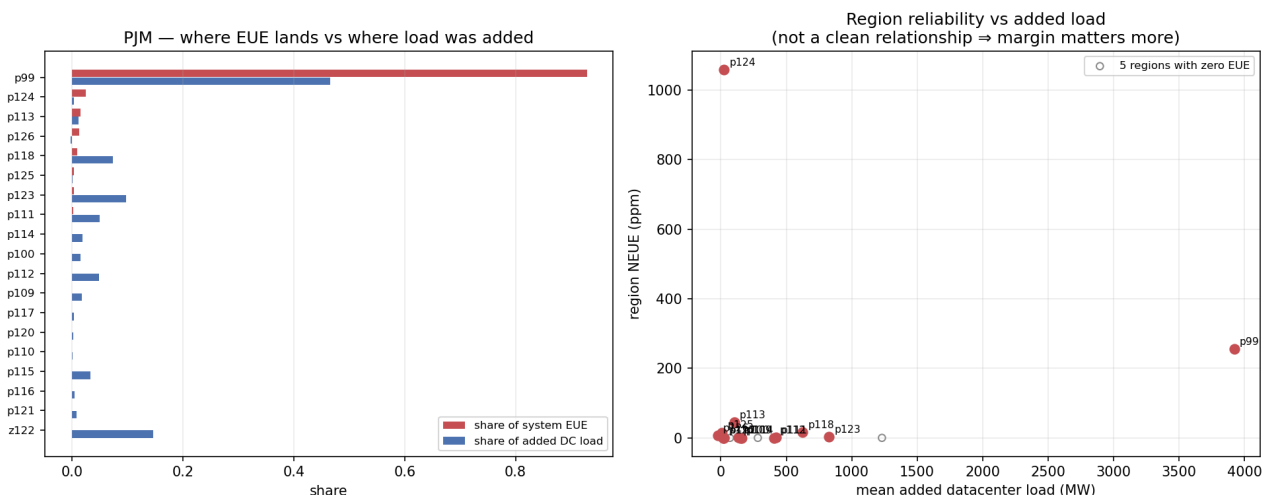


Figure 1: PJM where

PJM: 93 % of the unserved energy is in region p99, which also receives **47 % of the added datacenter load** — the most of any region. p99 runs a *negative* average margin (−7,798 MW;

its capacity is below its load on average), so the datacenter load lands squarely on the most capacity-short region.

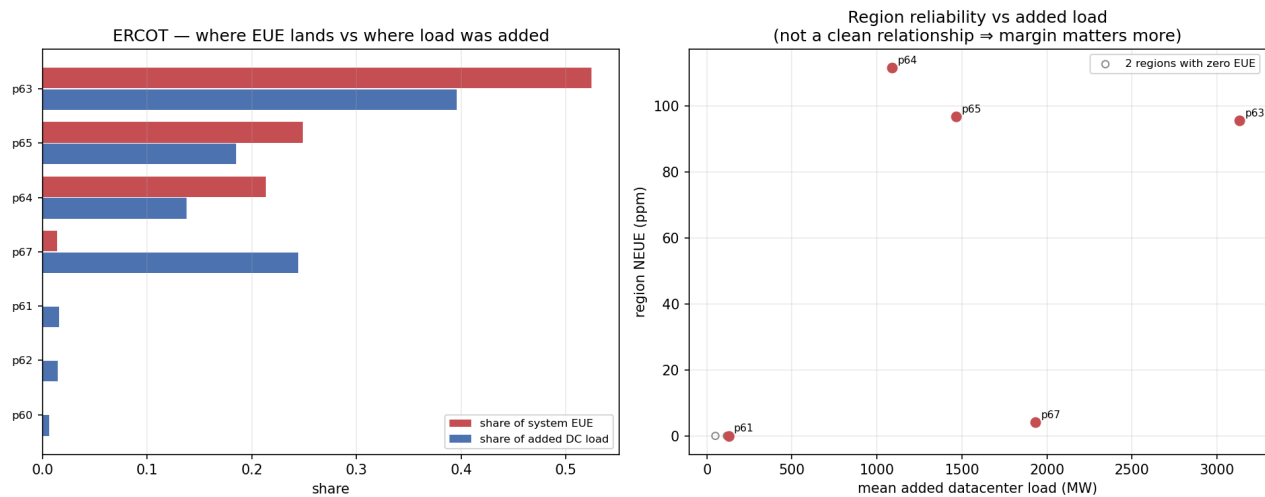


Figure 2: ERCOT where

ERCOT: 98 % of the EUE is in three regions — p63, p65, p64 (52 %, 25 %, 21 %), which together take 71 % of the added load. The one exception is instructive: **p67 receives the most added load (24 %) but produces only 1.4 % of the EUE**, because it carries enough margin to absorb it (minimum margin $-1,374$ MW vs $-5,000$ to $-15,000$ MW in the failing regions).

So in these systems datacenter DR is **well-targeted by construction** — it is sized to the load, and the load is where the risk is. Whether it is *sufficient* is a separate question (§6). This is the key reversal from the first-round runs, where EUE was concentrated in a region with almost no datacenter load.

3. The failures are system-wide capacity shortfalls, not local pockets

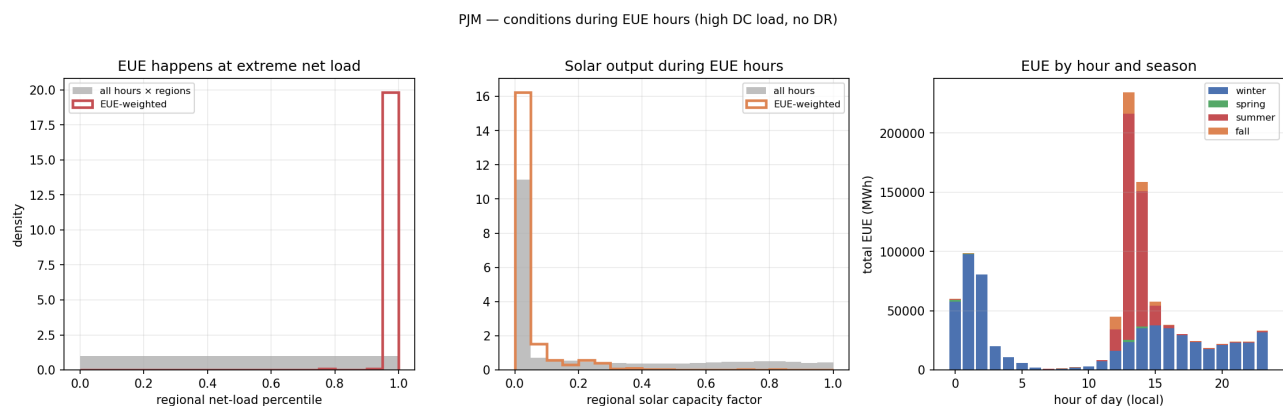


Figure 3: PJM conditions

condition (PJM)	all hours	during EUE (EUE-weighted)
regional net-load percentile	0.50	0.995

condition (PJM)	all hours	during EUE (EUE-weighted)
regional margin percentile	0.50	0.005
regional margin (MW)	+4,061	−17,464
solar capacity factor	0.227	0.035
wind capacity factor	0.425	0.331
added datacenter load (MW)	444	3,743

EUE occurs at the extreme top of net load and the extreme bottom of margin, in hours with **near-zero solar** (evenings and nights) and **elevated datacenter load**. The average margin during EUE hours is **−17 GW** — the system is short by many gigawatts, not by tens of megawatts. ERCOT is the same pattern (margin percentile 0.006, solar CF 0.02, EUE-hour added load 2,076 MW vs 1,130 all-hours).

Unlike the earlier runs, wind is **not** the trigger — wind CF during EUE hours is 0.33 (PJM) / 0.19 (ERCOT), only moderately below normal. The driver is total load against total capacity, with solar absent because the failures are after sunset.

4. Datacenter load creates as many new failure hours as it worsens

	base EUE	high EUE	share in <i>newly-failing</i> hours
PJM	16,477 MWh	1,003,700 MWh	52 %
ERCOT	10,168 MWh	804,376 MWh	54 %

About half of the added unserved energy falls in hours (and regions) that had **no** EUE in the base case. So the datacenter load is not merely deepening existing scarcity — it is creating new scarcity in previously-adequate hours. Consistent with this, the shape of *when* EUE occurs shifts substantially (total-variation distance base→high of 0.33–0.46 for PJM, 0.43–0.72 for ERCOT).

5. The problem is chronic, not a handful of extreme hours

case	EUE in top 10 h	in top 100 h	hours to reach 90 %	nonzero hours
PJM base	56 %	86 %	195	1,884
PJM +DC, no DR	10 %	44 %	627	7,855
ERCOT base	86 %	100 %	14	86
ERCOT +DC, no DR	12 %	66 %	211	1,023

In the base cases the unserved energy is a few extreme hours. Adding datacenter load **spreads it across hundreds to thousands of hours** — it takes 627 of the worst PJM hours to reach 90 % of the total. This is a sustained seasonal capacity deficit, which is exactly the kind of problem sustained DR can chip at.

6. DR helps materially in PJM, weakly in ERCOT — and is magnitude-limited everywhere

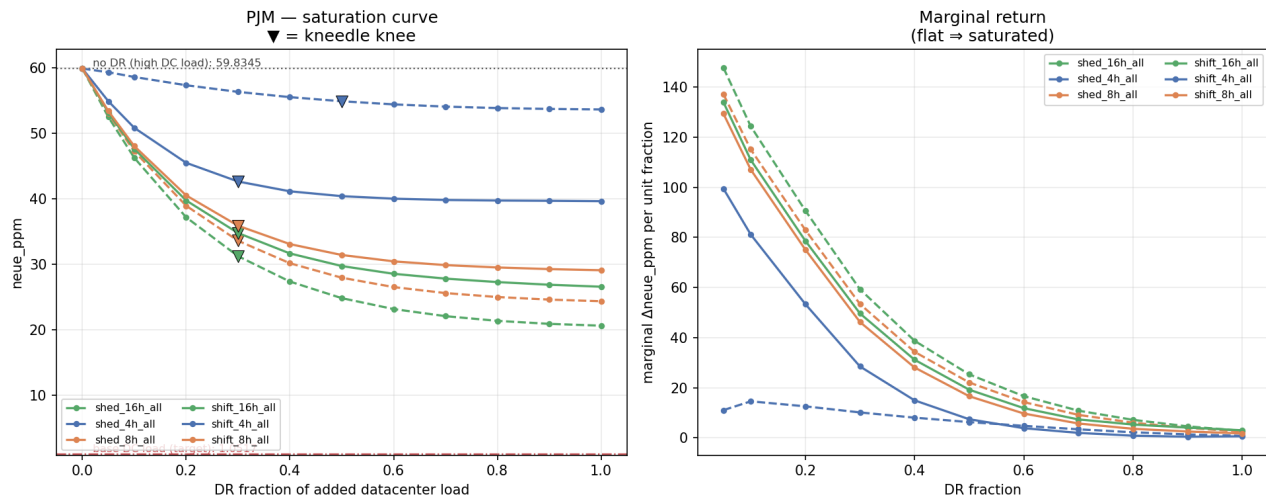


Figure 4: PJM saturation

DR family	system	NEUE floor at 100 %	share of DC penalty recovered
shed 16 h	PJM	20.6 ppm	67 %
shed 8 h	PJM	24.4 ppm	60 %
shift 16 h	PJM	26.6 ppm	57 %
shift 8 h	PJM	29.1 ppm	52 %
shift 4 h	PJM	39.6 ppm	34 %
shed 4 h	PJM	53.6 ppm	11 %
shift 8 h	ERCOT	56.2 ppm	22 %
shift 4 h	ERCOT	61.4 ppm	15 %

PJM's best case buys back **two-thirds** of the datacenter penalty — DR is genuinely effective here because it is well-targeted (§2) and the deficits, though large, are spread over many hours it can address. ERCOT recovers only ~**22 %**, but that is partly because ERCOT's sweep only includes **shift**, only **4 h and 8 h** — no shed, no 16 h. It is under-equipped in the experiment, not necessarily in principle.

Every family is magnitude-limited (outputs/tables/dr_limit_diagnosis.csv). The always-on shift devices are fully dispatched — PJM's shift_16h moves 5.9 million MWh — yet still leave a 383,000 MWh shortfall. DR is being used to the hilt; the system is simply short by more than the datacenter load can flex away. There is **no knee past which more DR is wasted**; returns decline smoothly and 80 % of the achievable reduction needs a fraction of ~0.4 (PJM) to ~0.7 (ERCOT).

Duration still beats deployment share. Within PJM, moving from 4 h to 16 h devices matters far more than raising the fraction — shed_4h recovers 11 % while shed_16h recovers 67 %.

7. The shed-vs-shift comparison remains confounded (PJM)

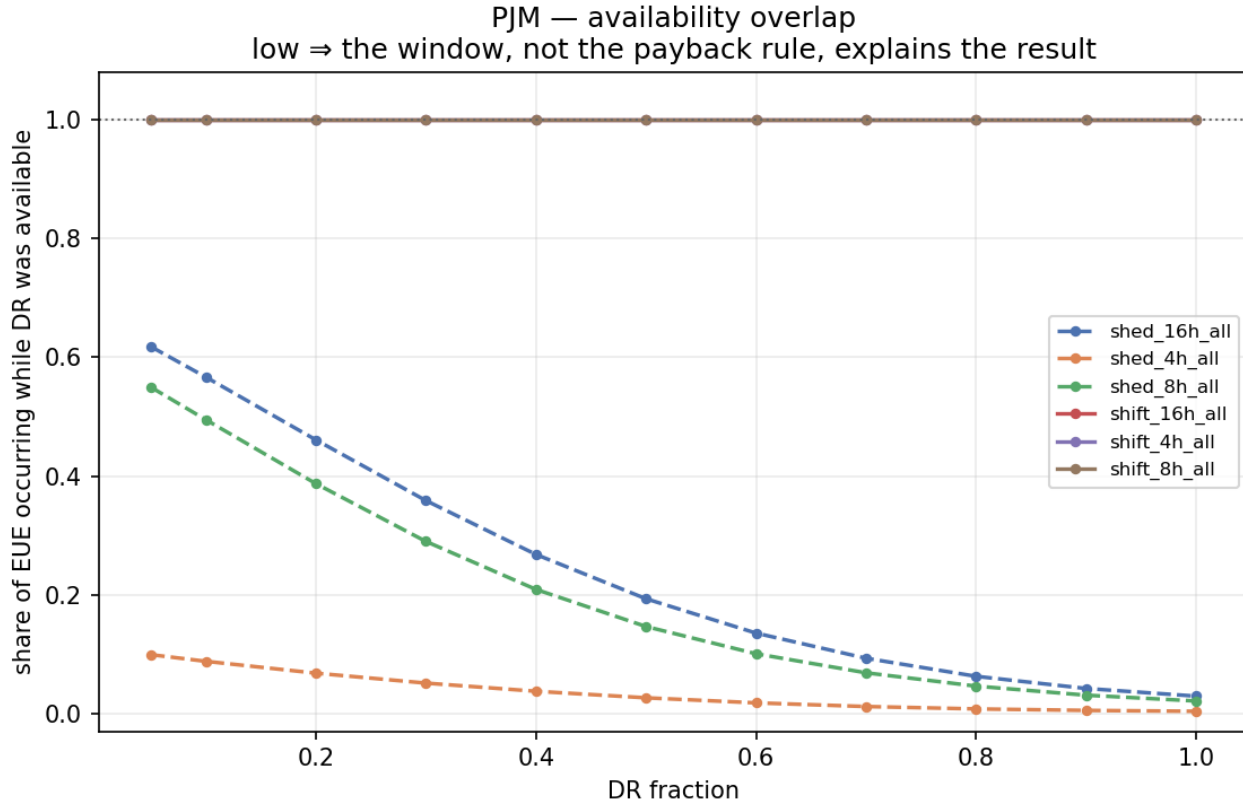


Figure 5: PJM availability overlap

As before, shed differs from shift in two ways at once — forgiven payback (an advantage) and a restricted daily availability window (a handicap). The window handicap is visible: shed_4h operates only 4–8 PM ET and only **0.4 %** of its remaining EUE falls in that window, which is why it is the weakest family despite shed’s mechanistic advantage. A clean comparison still needs an **always-on shed run** (drop available_hours_et) — a config change, not new code.

What this means

1. **Datacenter load, on this tighter grid, is a first-order adequacy problem in both systems.** NEUE rises 60–80× and the deficit is chronic and system-wide, averaging many GW short during failure hours.
2. **DR is well-targeted here and materially effective in PJM (–67 % of the penalty), but it cannot close the gap** because the shortfall exceeds what the flexible datacenter load can supply — every family is magnitude-limited. DR is a substantial mitigation, not a substitute for the generation that was not built.
3. **ERCOT looks worse than PJM partly because its DR toolkit in this sweep is thinner** (shift-only, ≤ 8 h). Whether ERCOT is genuinely harder to help or just under-equipped needs a shed / 16 h run before drawing a conclusion.
4. **Duration beats quantity.** Longer-duration flexibility is the stronger lever than a higher deployment fraction.

Caveats

- **Monte Carlo noise.** Case-level standard error is $\sim 1\%$ of EUE. With these much larger EUE values the DR-sweep increments are now comfortably resolved (the best PJM and ERCOT cases are flagged statistically distinguishable from their runners-up), but single-increment differences should still be read as indicative.
 - **Same buildout by construction.** Generation was not re-optimized for the higher load; these results describe datacenter load arriving *without* a supply response, which is the intended experiment, not a forecast.
 - **ERCOT's DR sweep is incomplete** relative to PJM's (no shed, no 16 h), so cross-system DR comparisons are not apples-to-apples.
 - **The DR device is regional, not datacenter-specific.** It is *sized* from datacenter load but PRAS reduces total regional load when it operates.
 - **Event counts depend on a threshold** (*_eps_sensitivity.csv); totals are robust, counts are not.
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Files

Output	Path
Figures (per system)	outputs/figures/{pjm,ercot}_{where,when,conditions}
Regional attribution	outputs/tables/{pjm,ercot}_where.csv
Timing / shape shift	outputs/tables/{pjm,ercot}_when.csv, _shape_shift.csv
Event conditions	outputs/tables/{pjm,ercot}_why.csv
New vs amplified failures	outputs/tables/{pjm,ercot}_delta.csv
Concentration / episodes	outputs/tables/{pjm,ercot}_concentration.csv, _events.csv, _eps_sensitivity.csv
DR saturation	outputs/tables/dr_saturation_{curves,summary}.csv, dr_limit_diagnosis.csv
Case registry	cache/registry.csv

Full detail, including data-quality issues found in the source files, is in [FINDINGS.md](#).